

The Path to 9.5+

NCRTC Bus Management System · Repository github.com/ltsAryanJ/intelligent-fleet-management-system · HEAD d36ea82

A code-grounded plan to lift the project from its v3 score of **7.6/10** to **9.5+** · Prepared 7 July 2026 ·
Derived from a 12-agent deep-dive across all four modules, auth, code, UI, docs, honesty and stretch

TARGET TRAJECTORY

7.6 → 9.6_{/10}

+2.0 · achievable in ~4
focused weeks

It's wiring and honesty, not new build.

The codebase already contains ~90% of every remaining must-have — a draft-inclusive notices list, a read-receipt roster, a geofence endpoint, an SLA calculator, a FileUpload component, a `/duties/roster` route, a depot-scoped WebSocket, an executive PDF — but leaves them orphaned, mis-wired, or fabricated in the UI. The distance to 9.5+ is closing dead-ends and telling the truth, not writing a new system.

THE ARITHMETIC IS FORGIVING

A maxed stretch bonus is worth ~+1.0 to the final /10, and **WebSocket (+3) + trip-replay (+2) = +5 are already banked**. So an overall 9.5 needs only a core-weighted ~8.6 *once* geofence (+3) and SLA (+2) are recovered. The three real risks: a geofence implementation contradiction that must collapse to a single `ST_Contains` + seeded-polygon + corrected-exit-breach path (radius circles are exactly what got DENIED); the absent dedicated Auth/RBAC track that otherwise caps Functional near 9.2; and the demo video being the tail dependency of every module fix — so it must be recorded last.

CATEGORY	WEIGHT	V3 NOW	TARGET	KEY MOVES THAT EARN IT	Δ /10
Functional	40%	7.1	9.2	Dedicated Auth/RBAC track (fix <code>GET /routes</code> leak, real logout, per-request re-validation) lifts Auth 7.5→9.3; CMS publish + read-receipt wired (5.5→9); server-side double-booking 409 + ordered-stops editor (Sched 7→9.4); photo evidence + notes-on-every-transition + continuous SLA (IMS 8→9.4); server-side depot filter + full side-panel (AVLS 8→9.3).	+0.84
Code quality	15%	6.8	9.3	Kill the password-leaking print + guard <code>SECRET_KEY</code> ; collapse 52 copy-paste depot-scope blocks into one shared dependency (gated behind a regression test) which also fixes the routes leak; <i>resolve</i> the geofence router (not delete); de-orphan FileUpload/StateDisplays; one exception envelope; a real Alembic baseline.	+0.38
UI/UX & mobile	15%	7.2	9.4	Installable PWA (manifest + service worker) with an offline driver duty view and a 320px bottom nav (44px targets, safe-area); wire the dead StateDisplays + add <code>isError</code> /retry branches; an accessibility pass (focus-visible, aria, keyboard-reachable cards, reduced-motion); expose the pagers where server paging already exists.	+0.33
Documentation	10%	6.8	9.5	Fix the fresh-clone Quick Start (<code>cd test1</code> → real dir); remove the fabricated <code>tests/</code> + phantom-Alembic claims; an honest module-status table + a "Real vs Simulated vs Seeded" section + a Demo→Production delta appendix; embedded	+0.27

CATEGORY	WEIGHT	V3 NOW	TARGET	KEY MOVES THAT EARN IT	Δ /10
				screenshots; a right-sized architecture note reconciled to HEAD.	
Demo & presentation	15%	7.0	9.4	A fail-loud seed so a green container <i>means</i> a demoable system; a flawless <code>git clone && docker compose up</code> ; a scripted 10-min <code>DEMO.md</code> mapped to the PRD flow; recorded last , after every module fix lands, so no beat is a dead-end.	+0.36
Honesty & judgment	5%	6.0	9.5	Render the copilot demo-mode badge + wire <code>response.mode</code> to <code>settings.COPILOT_MODE</code> (structural, can't desync); a truthful logout; delete the password print; correct the "real-time tracking" overclaim; a committed no-fabricated-states audit; a Demo→Prod slide that <i>weaponises</i> transparency.	+0.18
Stretch (bonus)	≤+10%	6.5	9.5	Bank WebSocket + trip-replay (+5) with regression tests; recover Geofence DENIED→FULL (+3) via one seeded-polygon <code>ST_Contains</code> + exit-breach + map/banner consumer; SLA PARTIAL→FULL (+2) via a 60s persisting sweeper + computed-on-read.	+0.30

Weighted math: core categories → $(9.2 \times .40) + (9.3 \times .15) + (9.4 \times .15) + (9.5 \times .10) + (9.4 \times .15) + (9.5 \times .05) \approx 9.32$, plus a near-maxed stretch bonus $\approx +0.28$ incremental → $\approx 9.6 / 10$. Even in the conservative case where geofence lands only partial, the banked +5 and SLA→FULL still clear 9.5.

- 1 Reconcile geofence to ONE implementation** and recover the DENIED +3: seed `Depot.geofence` polygon, swap haversine for `ST_Contains`, fix `Route.stops→route_stops`, flip to the exit-breach condition, then wire the map-polygon + banner consumer.
- 2 Fix the CMS create→publish dead-end:** point `NoticesPage` at `GET /notices` with Draft/Published tabs so drafts appear and the existing Publish button renders — unblocks the module's headline flow and the demo climax.
- 3 Stand up the absent Auth/RBAC track:** fix the `GET /routes ?depot_id` cross-depot leak, add per-request permission re-validation (or delete the 3 dead RBAC tables), token-denylist logout, `SECRET_KEY` guard, delete the password print.
- 4 Add the 60s SLA sweeper** that persists `sla_breached` + computed-on-read: SLA PARTIAL→FULL (+2).
- 5 Wire the orphaned read-receipt roster** `GET /notices/{id}/readers` with server-resolved recipients and an N-of-M progress bar (a MUST-HAVE with zero callers today).
- 6 Render the copilot demo-mode badge** and wire `response.mode` to `settings.COPILOT_MODE` — kill the hardcoded "Online" fabrication that is fatal on stage.
- 7 Write the real pytest suite on Postgres+PostGIS** (auth, RBAC, depot-scoping RED-first, incident state machine, notice targeting, geofence, SLA) + a GitHub Actions CI that runs it — replaces the fabricated `tests/` claim.
- 8 De-orphan FileUpload:** fix the uploads route-order bug and wire photo evidence into the IMS drawer and attachments into notices.
- 9 Ship the ordered-stops / planned-arrival route editor** (drag-reorder, per-stop offsets) + server-side double-booking 409 + wire `/duties/roster` — completes Scheduling and unblocks AVLS on-time/late.
- 10 Make the fresh clone flawless** (README `cd` path, fail-loud seed, Alembic-or-honest-docs) and record the scripted 10-min demo video **last**, after every upstream fix lands.

P0 · Fix the dead-ends & correctness 3–4 DAYS

→ 8.4

Every headline flow reaches its endpoint; every glaring security / fabrication smell gone; geofence reconciled to one honest implementation. No must-have is a dead button after this phase.

L Collapse geofence to ONE implementation — seed `Depot.geofence` `POLYGON + ST_Contains` + corrected exit-breach (recovers DENIED +3)

seed.py:169 (only sets radius, polygon NULL) · geofence/router.py (fix Route.stops→route_stops L242, replace haversine with ST_Contains, flip to own-vehicle-left-home-polygon) · models.py:177

IMPACT Geofence DENIED→FULL (+3, the single biggest lever); Code +0.3 as an orphaned+broken router becomes live.

S Fix the CMS create→publish dead-end — split `NoticesPage` onto `GET /notices` with Draft/Published tabs + a working Publish

NoticesPage.tsx:27 queries /notices/feed so drafts never appear and Publish L113–120 never renders · backend GET /notices + POST /notices/{id}/publish already work

IMPACT CMS 5.5→~7.5; unblocks the module's headline PRD flow and the demo climax.

M Dedicated Auth/RBAC hardening — fix the `GET /routes` cross-depot leak, guard `SECRET_KEY`, kill the password print, real logout

routes/router.py:116–119 (scope applied only when no depot_id param → driver passes ? depot_id=other) · config.py:35/96 · auth/router.py:44–47 (no-op logout claims success)

IMPACT Auth 7.5→8.5; Honesty (removes secret leak + false success). The absent track the critique flagged as the biggest weighted drag.

S Render the copilot demo badge + wire `response.mode` to `settings.COPILOT_MODE`

CopilotPage.tsx:26/34 (copilotMode captured, never rendered) & L89 (hardcoded 'Online') · copilot/router.py stamp resp.mode

IMPACT Honesty 6.0→~8; kills the one fabrication fatal on stage. Structural, so it can never desync.

M Fix the uploads route-order bug + wire FileUpload into the IMS incident drawer (photo evidence)

uploads/router.py (move /list above /{file_id}) · FileUpload.tsx L28 (wrong list URL) · IncidentDetailDrawer.tsx (zero importers today)

IMPACT IMS photo-evidence must-have absent→done; de-orphan FileUpload (Code).

M Server-side double-booking prevention (409) + wire `/duties/roster`; fail-loud Dockerfile seed; fix README `cd test1`

duties/router.py create_duty+bulk_assign (add _assert_no_conflict) · RosterPage N+1 → /duties/roster · Dockerfile:27 (|| echo swallows failure) · README.md:128

IMPACT Scheduling double-booking Partial→Full; Demo (green container = demoable); Docs first command works.

Every PRD must-have across all four modules + Auth is wired, enforced and non-orphaned; continuous SLA; complete side-panels.

M Continuous SLA sweeper (60s) persisting `sla_breached` + computed-on-read

incidents/router.py:481 (breach only stamped at resolve) · `_incident_to_dict`:608 · main.py lifespan (add `sla_sweeper` task like `gps_simulator`)

IMPACT Stretch SLA PARTIAL→FULL (+2); the badge finally fires on live overdue incidents.

M Wire the read-receipt roster `GET /notices/{id}/readers` — recipient-resolved N-of-M + `acknowledgement_required` column

notices/router.py:312 (orphaned, no denominator) · NoticeDetailDrawer.tsx (only aggregate count today) · models.py Notice (+`acknowledgement_required`)

IMPACT CMS read-receipt MUST-HAVE Partial→Full; de-orphans the endpoint.

L Ordered-stops / planned-arrival-times route editor (drag-reorder + per-stop offsets) consuming the dead `RouteStopAssign` schema

routes/router.py (add `PUT /{id}/stops`) · new RouteStopsEditor.tsx using `framer-motion` Reorder (already a dep)

IMPACT Scheduling marquee must-have; also unblocks AVLS real on-time/late detection.

M Complete AVLS — server-side depot filter param + side-panel driver/route/last-30-min polyline; require notes on EVERY IMS transition

AVLSPage.tsx (passes zero params to `/gps/live`, client-side filter L156) · gps/router.py (today's-Duty join + `/gps/live-trail`) · incidents/router.py `acknowledge/assign/start/close` (notes optional today)

IMPACT AVLS side-panel + scoping must-haves; IMS audit-trail accountability.

M Per-request permission re-validation (close the JWT-baked-permissions weakness) OR delete the 3 dead RBAC tables + route delete + copy-week

dependencies.py:55 `get_current_user` · models.py:189–229 (Role/PermissionModel/RolePermission never queried) · routes/router.py (guarded `soft-delete`) · duties (copy-week)

IMPACT Closes the last Auth defect; retires scope bloat; completes Scheduling CRUD.

M Wire the dead StateDisplays + add missing `isError` /retry branches across 6+ list pages

shared/StateDisplays.tsx (zero importers) · IncidentsPage/NoticesPage/DutiesPage/RosterPage/UsersPage/VehiclesPage (raw skeleton, no error branch)

IMPACT UI consistency; removes the dead-component penalty; failures stop looking like empty screens.

Real tests + CI proving every invariant; the stretch bucket maxed and protected; a11y + PWA polish; one canonical error / scope / schema mechanism.

L Real pytest suite on Postgres+PostGIS — auth, RBAC, depot-scoping, incident state machine, notice targeting, geofence `ST_Contains`, SLA + trip-replay

new backend/tests/ + conftest with a postgis service · write the depot-scoping test RED FIRST to guard the routes-leak fix and the 52-site refactor

IMPACT Code +0.8, Honesty +0.6 (replaces the fabricated tests/ claim); protects every stretch claim.

M GitHub Actions CI (postgis service) — backend pytest + ruff + frontend build/tsc/vitest on every push

new .github/workflows/ci.yml (none exists at HEAD) · README CI badge

IMPACT Code +0.7, Honesty +0.4; a green badge on a fresh clone.

L Collapse 52 copy-paste depot-scope blocks into one shared dependency, GATED behind the now-green depot-scope test

new core/dependencies.scope_to_depot · 13 routers · sequence AFTER the regression test exists

IMPACT Code +0.4; unbypassable scoping. Do NOT ship blind pre-demo.

L Installable PWA (manifest + SW) + offline driver duty view + 320px bottom nav + a11y pass

index.html · vite.config.ts · main.tsx · AppShell.tsx · DriverDashboard.tsx · index.css

IMPACT UI +0.8 (credit to UI/Demo, NOT the stretch axis); the mobile-first bar + a11y examiner probe.

L Alembic baseline capturing the new CMS columns + geofence polygon; keep `create_all` as the Docker path until proven on a fresh PostGIS clone

backend/alembic/ (phantom today) · Dockerfile CMD (do NOT flip to upgrade head until geoalchemy2 Geometry autogen verified)

IMPACT Code +0.4, Honesty +0.5; removes phantom-Alembic. Must include acknowledgement_required / title_hi / content_hi or a fresh clone drifts.

M Standardise on the custom-exception envelope (zero raw HTTPException); unify geofence/SLA/route-deviation over the existing broadcast as one Ops-Alerts stream

uploads/router.py (8 raw HTTPException) · copilot inline import · websocket.py envelope + OpsAlertsToaster

IMPACT Code +0.2; makes the WebSocket stretch visibly load-bearing across 3 modules, not just dot-moving.

Fresh-clone perfection, screenshots, honest docs, and a scripted demo video recorded LAST once every upstream fix has landed.

M Screenshots + a scripted 10-min DEMO.md mapped click-for-click to the PRD flow, recorded LAST

new docs/screenshots/ · DEMO.md · the CMS publish + read-receipt climax and geofence/SLA beats depend on P0–P2 landing

IMPACT Demo +1.2 (the largest Demo lever); no beat may be a dead-end per its own acceptance test.

M Honesty docs — "Real vs Simulated vs Seeded" table, honest module-status column, a Demo→Production delta appendix/slide, beyond-PRD tables justified

README.md · ARCHITECTURE.md · reconcile the absolutist "verified against implementation" to "reflects HEAD @ <commit>"

IMPACT Honesty +0.7 (top-of-band, self-drawn demo/prod line), Docs +0.3.

S Enrich the already-built executive PDF with the new geofence/SLA KPIs + a one-click board-pack button; commit a final no-fabricated-states audit

reports/pdf_generator.py (generate_executive_pdf already exists) · ExecutiveDashboard.tsx

IMPACT Demo +0.3, Docs +0.3; leverages sunk cost; proves the stretch features feed a real artifact.

WebSocket +3 · banked, protect with tests

Trip replay +2 · banked, protect with tests

Geofence +3 · DENIED → recover FULL

SLA +2 · PARTIAL → FULL

Bank the +5 already earned by protecting it with regression tests: a `/gps/stream` connect/close smoke test and a `/gps/history` test asserting ordered pings and that the >200km/h implausible-jump filter fires — otherwise a silent regression quietly costs the +2.

Recover Geofence (DENIED → FULL, +3) with exactly ONE implementation — the plans contradicted each other and the radius-circle path is precisely what got denied: (1) seed `Depot.geofence` as a real `POLYGON(srid=4326)` (today only `geofence_radius_m` is set, so the polygon column is NULL and every `ST_` query returns empty); (2) replace the haversine circle test and the broken `Route.stops` reference with `ST_Contains(Depot.geofence, ST_SetSRID(ST_MakePoint(lon, lat), 4326))`; (3) flip the breach condition to the PRD's real requirement — a vehicle that has *left* its home-depot polygon while off-shift, plus unauthorised entry — not the inverted "inside a foreign depot" check; (4) build the frontend consumer: draw depot polygons from `ST_AsGeoJSON` and a live red breach banner over the existing WebSocket, so a grader visually watches the fence get crossed.

Take SLA (PARTIAL → FULL, +2) with a 60s lifespan sweeper that persists `sla_breached` on live overdue incidents plus a computed-on-read fallback so the badge is correct between sweeps. Route the new geofence/SLA/route-deviation events through the single existing `gps_manager` broadcast so WebSocket reads as load-bearing across three modules, not just dot-moving on a map. (PWA and the executive PDF are genuine "wow" but add zero to the capped stretch axis — credit them to UI / Demo only, and build the PWA once.)

05 DEFINITION OF DONE — THE 9.5+ CHECKLIST

- ✓ **Fresh clone:** `git clone && cd ... && docker compose up --build` brings up three healthy containers and loads the login page with copy-paste-from-README commands and zero edits.
- ✓ **Every must-have end-to-end for its role:** AVLS live+history+depot-scope+full side-panel; Scheduling routes (incl. delete) + ordered-stops/arrival editor + roster publish + *prevented* double-booking (409); IMS create→ack→assign→in-progress→resolve→close with notes on every step, photo evidence and a continuously-correct SLA badge; CMS author→draft→publish→feed→read→acknowledge→read-receipt roster.
- ✓ **Zero dead / orphaned / fabricated code:** no zero-importer components (FileUpload, StateDisplays wired), no orphaned endpoints (`/notices/{id}/readers`, `/duties/roster`, `/geo/*` all have callers), no dead state driving a false label (the copilot badge is data-driven), no broken paths (geofence `Route.stops` fixed and `ST_Contains` -backed).
- ✓ **No security / honesty smells:** no DB password in logs, `SECRET_KEY` refuses the dev default outside dev, `GET /routes` cannot be coerced cross-depot, logout makes no claim it doesn't fulfil, and the copilot is honestly labelled demo mode.
- ✓ **Real tests + green CI:** pytest on Postgres+PostGIS proves auth, RBAC, depot-scoping, the incident state machine, notice targeting, geofence `ST_Contains` and SLA recompute; frontend vitest covers the auth store / PermissionGuard; GitHub Actions runs both on every push with a linked badge.
- ✓ **Stretch maxed and protected:** WebSocket + trip-replay banked with regression tests, Geofence FULL via seeded-polygon `ST_Contains` + map/banner, SLA FULL via the persisting sweeper — arithmetically ~+1.0 to the final /10.
- ✓ **Docs match reality:** no fabricated `tests/` or phantom-Alembic claims, honest "Real vs Simulated vs Seeded" + module-status tables, embedded screenshots, a Demo→Production delta appendix, a right-sized architecture note reconciled to HEAD.
- ✓ **A scripted 10-minute demo video, recorded last,** walks the full PRD flow click-for-click with no dead-end at the CMS-publish / read-receipt climax or the geofence / SLA beats.

- **The geofence contradiction is the single biggest swing lever.** If the radius-circle path ships it stays DENIED (it is exactly what a grader rejected as "wrong condition / orphaned") and the frontend gets wired to the wrong contract. **Mitigation:** reconcile to the seeded-POLYGON + `ST_Contains` + exit-breach path *before* building the banner/polygon consumer.
- **No owner for Auth/RBAC end-to-end caps Functional at ~9.2–9.3** (Auth is a 7.5 component of the 40% axis). **Mitigation:** the dedicated P0/P1 Auth track; if it slips, overall still clears 9.5 on stretch + Docs + Honesty but with thin margin.
- **New CMS columns + the geofence polygon rely on runtime `create_all` and are in no migration.** Flipping the Dockerfile to `alembic upgrade head` before the baseline captures them boots a fresh clone *without* those columns. **Mitigation:** author the baseline AFTER those columns land, and keep `create_all` as the Docker path until the migration is proven on a fresh PostGIS clone.
- **The 52-site depot-scope refactor is high regression risk across 13 routers immediately pre-demo** — and it is the mechanism that fixes the routes leak. **Mitigation:** land the RED-first depot-scoping regression test before the refactor and gate the refactor behind it; do not ship blind.
- **The demo video is the tail dependency of the entire programme** — its CMS climax and geofence/SLA beats require every module fix landed. **Mitigation:** record LAST, treat "no dead-end" as its acceptance test, keep P0 CMS-publish + P1 read-receipt on the critical path.
- **Alembic Geometry-column autogenerate routinely produces a non-applying baseline** (geoalchemy2 pulls in spatial system tables) — effort is realistically L, not M. **Mitigation:** hand-edit the baseline to `CREATE EXTENSION postgis` first and verify on an empty PostGIS DB before touching the Dockerfile.
- **AVLS real on-time/late detection depends on Scheduling's planned-arrival offsets.** If that link is never wired, both ship an honest speed-only heuristic and each leaves ~0.4 on the table. **Mitigation:** sequence the stop-editor before AVLS operational colouring, or explicitly label "Idle/Moving" to preserve honesty.

HOW THIS ROADMAP WAS BUILT

Twelve agents each read the actual source at HEAD d36ea82 and produced concrete, file-level improvement plans across the four modules, auth, code quality, UI/UX, docs, honesty and stretch; an adversarial critic then pressure-tested whether executing all of it clears 9.5 under the exact rubric weights and corrected the cross-dimension sequencing; a synthesis pass built the scoring model and phased plan. Effort sizes: S < half a day, M ~1–2 days, L ~3–5 days. Scores are calibrated to the same MVP-depth Lite-PRD bar used in the v1→v2→v3 evaluations. The plan deliberately favours wiring and fixing existing backend work over new build — which is why +2.0 is realistic in roughly four focused weeks.